📝 Unit 3 Classwork — Equations & Inequalities

One/Two-Step · Distributive · Both Sides · Multi-Step · Special Cases · Inequalities · Word Problems

A. One- & Two-Step Equations

Problem 1

Solve: $x + 9 = 15$

6

Problem 2

Solve: $4x = 28$

7

Problem 3

Solve: $3x + 5 = 20$

5

Problem 4

Solve: $\frac{x}{3} - 2 = 4$

18

Problem 5

Solve: $-2x + 7 = 1$

3

B. Distributive Property & Variables on Both Sides

Problem 1

Solve: $2(x + 3) = 16$

5

Problem 2

Solve: $3(2x - 1) = 15$

3

Problem 3

Solve: $5x + 4 = 2x + 13$

3

Problem 4

Solve: 7x − 2 = 3x + 14

4

C. Multi-Step Equations

Problem 1

Solve: 3(x + 2) = 2x + 10

4

Problem 2

Solve: 4x − 3 = 2(x + 5)

6.5

Problem 3

Solve: 5 − 2x = 3x + 15

−2

D. Special Cases — No Solution or Infinite Solutions

Problem 1

Solve: 4x + 3 = 4x + 3 — how many solutions?

infinitely many solutions

Problem 2

Solve: 2x + 5 = 2x − 1 — how many solutions?

no solution

Problem 3

Solve: 3(x + 4) = 3x + 12 — how many solutions?

infinitely many solutions

E. Linear Inequalities

Problem 1

Solve: x + 4 > 9

x > 5

Problem 2

Solve: $3x \leq 21$

$x \leq 7$

Problem 3

Solve: $-2x > 8$ (careful!)

$x < -4$

Problem 4

Solve: $5x - 3 \geq 12$

$x \geq 3$

Problem 5

Solve: $-3x + 1 \leq 10$ (careful!)

$x \geq -3$

F. Word Problems

Problem 1

A taxi charges $3 plus $2 per mile. If the total fare was $17, how many miles were driven?

7

Problem 2

A gym membership costs $25 to join plus $15 per month. You can spend at most $100. What's the max number of months?

5